

Mine in a box™

Managing Operations

Part A · Supervisor Handbook

Systems that make good decisions inevitable.

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1. Welcome to the program

Managing Operations runs a mine's production from the shift up. It starts with you: the person who runs the shift.

Your shift is five stepping stones, crossed in order: the handover, the pre-start, time in the field, time in the office, and the handover to the next crew.

Section 1 · A letter from Dawid

Section 2 · How the program runs

A letter from Dawid

Welcome! For too long I have watched supervisors get promoted, handed a radio, some hope and promises of great things to come, and then left to fend for themselves.

In my experience, many, if not most, of you spend most of your time battling chaos and poor plans, chasing targets you've had no say in, and then get blamed when things go wrong. If you did get training, it was usually some well-meaning leadership course that had little to do with what you are expected to do every day.

I designed this program with the help of successful supervisors from across Australia, and crusty old mine managers who started in mining when breakfast was a cigarette and a good look around, to help you get control of your day back. It clearly details where your job and accountability start and stop. It will also prepare you to become a Superintendent, should you choose to go that route.

Very few things in this book are new. Most of it is well-structured common sense, and the things and processes proven to work in the pit. Thank you for what you do.

– Dawid Falck



How the program runs

The program runs over one quarter, mostly on your own shift, with your own crew, coached by your own Superintendent.

Step	When	What happens
Before you start	Weeks 1 and 2	Before you start, your Superintendent starts their own program, because, like you, they are going to be taught how to run a day that helps you run yours, and because they will be coaching, teaching and assisting you through this program.
Prime	Weeks 3 to 5, on shift	Before you come to class, there are a number of activities you need to complete in a workbook. They are not complicated and won't take forever to do: they are designed for you to do in the workflow over one roster. We do this because I want to hear from you when we get together. Your experience and knowledge are valuable and need to be respected. Don't just rush through them; think about what they ask. This is your opportunity to provide input, because we can change things to fit your work if we need to! Remember: rubbish in, rubbish out.
The Validation Lab	Week 6, one day	The Validation Lab is where we get together for one day to review your Prime workbook and then run through this handbook: the five stepping stones and which decisions are yours. You get this handbook and the Apply tools at the Lab, not before.
Apply	Weeks 7 to 12, every shift	You run the shift to the cards. Your Superintendent will coach you as you use the tools in the handbook. You will be required to fill in the coaching cards for each shift, meeting and so on, as you self-evaluate your progress. This is where the new way of working becomes a habit.
Verify	Week 13	One shift observed, Yes or No against the cards.

"In this program we are teaching you how to manage. Management is the structuring of work and decisions so that performance remains predictable under pressure." – Dawid Falck

Where the learning happens (share of the program)



Continuous Mastery Loop™



2. What is work for you?

This chapter sets out your work: your job, where it stops and your Superintendent's starts, how to manage, the decisions placed before the work, the assumptions your shift was priced on, and the numbers you run it on.

Section 1 · Your job

Section 2 · Your job versus your Superintendent's job

Section 3 · The definition of management and the six behaviours

Section 4 · Let's talk about decisions

Section 5 · Let's talk about assumptions

Section 6 · Know your numbers

Your job

You run the shift. You own the minutes, the hours and the days: this shift and the next. The decisions that come before the work (the plan, the targets, the rules, the contingencies) are placed for you, by your Superintendent and the site.

You should give input into these rules and decisions, and you should be aware of the plan for the month and the targets, but they are not your responsibility. You can only work with what you have been given. It is your responsibility, though, to feed back to your manager the realities and risks on the ground, in the pit, and to make recommendations so that the decisions they make are based on fact.

You are accountable for the implementation and execution of the work allocated to you in your roster and shift, and for setting up the next shift for success. You are also accountable for the quality of the pit's work areas, the safety of your crews, and the coaching and upskilling of the operators in your crews, with the help of training, so that the way work is done is to standard. You are accountable for making sure the pit runs as a smooth, well-structured unit. You are not here to be a glorified cab driver, a glorified leading hand or a semi-professional firefighter. If this is your current work experience, it needs to stop.

Most importantly, you are **not** responsible for delivering unattainable targets. You are responsible for the effective and profitable use of your people and machinery, making sure your operators perform to standard. A poorly structured shift, slow operators, a long pre-start, first and last buckets not protected, sloppy lunch breaks: these are on you. Targets set without any consideration of the reality of the pit are not your problem. You will see in the four-block model where a mechanism exists for you to manage this.

Your shift reports and shift handover are both your protection and your opportunity to help your Superintendent, by giving them the information and recommendations they need to make better decisions and influence up.

Last point: when plans keep changing and you keep getting interrupted, the work cannot flow. The process for running a shift that follows is designed to protect you. If anyone issues orders that do not match the agreed plan or the four-block process, you now have the ability and the responsibility to say, "I'm sorry, this is not how we work around here", and, if you are forced to comply, to escalate it to your Superintendent's manager.

This is not a training program for the fun of it. This is how your site has agreed to work. Trust the process. Protect the rhythm of work!

Your job versus your Superintendent's job

Your job is **not** the same as your Superintendent's. You have two very different jobs running in parallel. Your Superintendent is responsible for creating the system you work in. They set the targets and the golden rules (with your input) and focus on the weeks and months. They work with the other Superintendents (drill and blast, maintenance and so on) to make sure all the bigger systems of work link up smoothly.

They do not get to fiddle with your shift and your people. If they have an issue, they talk to you. You don't get to work on the plans for the weeks to follow. Your job is this shift, this hour, the minutes. Respect flows both ways, and it starts with clarity on what is and what is not work for each of you.

Levels of work	Supervisor · level II	Superintendent · level III
Time span	3 months to 1 year.	1 to 2 years.
Your focus on shift	This shift and the next.	The day across crews, the week and the month.
Accountability / output	The number delivered or explained; an honest Shift Report.	A system that works; supervisors who grow; the department's number.
Decides	Who does what, where. What to correct. Which contingency to trigger. The number committed at the Drive-around. Stopping unsafe work. Any worker may. What escalates, and when. Continue, correct, prepare or escalate at the end of each block.	The week's plan. Rosters and crews. Standards, TARPs and golden rules. Spend. Which live decisions become rules.
Does not decide	The plan and targets. Rosters. Hiring. Standards. Spend.	How the shift is run. Anything inside the supervisor's authority.
The test	A fix inside this shift, within a contingency, is yours. Anything else goes up.	A decision re-made upstairs means a rule is missing.
Where the time goes	At least 90% in the field.	Checking, coaching and fixing what keeps coming back.
Decisions before the work	Finds the decisions made live, again and again.	Turns them into rules and checks them at Block 1.
Sign-offs	Fills in the cards, self-evaluating each line Yes or No, and the Shift Report.	Coaches the supervisor on them, every shift.
Coaching	Is coached; coaches one thing per operator per shift.	Coaches each supervisor; one point, checked next time.
Assumptions	Says when the ground does not match the plan.	Checks the site's assumptions against actuals each week.

In the field. Spend at least 90% of your shift in the field. The office is for the handover, the Shift Report and what cannot be done anywhere else.

What you must know. What a good digger operator and a good dozer operator look like. What a clean face and a good pit floor look like. What an effective circuit looks like: you must be able to spot when it has too many trucks, too few trucks, or a mixed fleet that is slowing everything down. Remember, being slightly under-trucked is always better than being over-trucked. But this depends on the digger operator being effective.

The definition of management and the six behaviours

> **Management is defining work, allocating work, coordinating work and verifying work.**
>
> **Management is the structuring of work and decisions so that performance remains predictable under pressure.**

#	Behaviour	Do I...?	Core function
1	Plan	Do I know the target, the risks and the controls before the shift starts?	Define
2	Assign	Does every person leave pre-start knowing their task, machine, area and target?	Allocate
3	Watch	Do I see the work with my own eyes, in the field, every block?	Coordinate
4	Respond	Do I tell people straight away what was good and what must change, and help them do it better?	Coordinate
5	Fix	Do I find and remove the cause of a problem, not just the symptom?	Coordinate
6	Report	Do I report my shift in PAVA (Plan, Actual, Variance/Impact, Actions taken) and set up the next shift for success?	Verify

What the research found. The best managers spend more time watching the work, so the feedback they give matches what people really did (Komaki; see "The evidence behind the model", page 45). That is why this program keeps you in the field 90% of your shift: you cannot respond well to work you have not seen.

Time in the seat. Every roster, spend two hours doing an operator's job, in the truck or the digger seat, and write down one thing you noticed.

The six behaviours. Each roster you rate yourself Yes or No on the six behaviours. Each is rated two ways: what you do on shift, and what you instil in your crew so it happens when you are not there. You and your Superintendent agree one behaviour to work on, and what they will look for in the field. Both sign. Use Apply Tool 8, the six behaviours self-evaluation.

Managing safety: running a high-reliability shift

A high-reliability organisation (HRO) does dangerous work every day and still has very few serious failures: aircraft carriers, air traffic control, nuclear power plants. They do not get there by luck or by more rules. They get there by how their people think and act every day, on every shift.

This does not replace your site's safety program. It strengthens it.

Principle	The question you ask	Where in your shift
Preoccupation with failure	"What did we get away with yesterday?" Write down the answer and follow it up.	Step 2, pre-start
Sensitivity to operations	"What is really happening at the work right now?" Engage the operator at every work area; each roster, two hours doing an operator's job.	Step 3, Blocks 1 to 4 in the field
Reluctance to simplify	"Why?" five times after anything we got away with. Stop at the system cause, not the person.	Step 3, Block 3 (or straight after an event)
Deference to expertise	"Who is closest to this, and what do they say?" The person who knows decides. A stop-work is always backed.	Any block, any decision on conditions
Commitment to resilience	"What nearly went wrong today, and has it been handed over?" Keep it open until fixed.	Step 4 Shift Report and Step 5 handover, in Block 4

Your Superintendent checks these five with you once a roster, Yes or No, with evidence.

Your crew

Your crew is your responsibility: not just getting them to their gear on time or making sure they keep to lunch breaks, but making sure they are effective as a team. Operators starting all the trucks up after lunch means the last truck idles for almost 10 minutes before it gets to move off, burning fuel and SMU hours unnecessarily. It is your responsibility to make sure they all work effectively, and to teach them how to work as a team. Focus on the small things and the big things will look after themselves: for example, teaching the truckies not to all take the same line around a corner into a work area, which creates a divot that stresses tyres and shocks. These small things may seem irrelevant, but running a tight ship is the secret to success.

Story time: broken windows

In the early 1990s, New York was out of control: crime was high, the city was dirty, run down and close to broke, and people and businesses were leaving. Rudy Giuliani was the 107th Mayor of New York, from 1 January 1994 to 31 December 2001 (elected in 1993 and again in 1997).

By mid-1998, overall reported crime was down 49.3% and homicides were down 69.3% (first half of 1998 vs 1993).

By the end of 2001, major crime was down about 62%, murders were down 66% and shootings were down about 74%, and New York had the lowest crime rate of America's largest cities, on FBI figures. He inherited a $2.3 billion budget gap, turned into a surplus of about $2.9 billion by 2001.

What turned it around? His answer was broken windows.

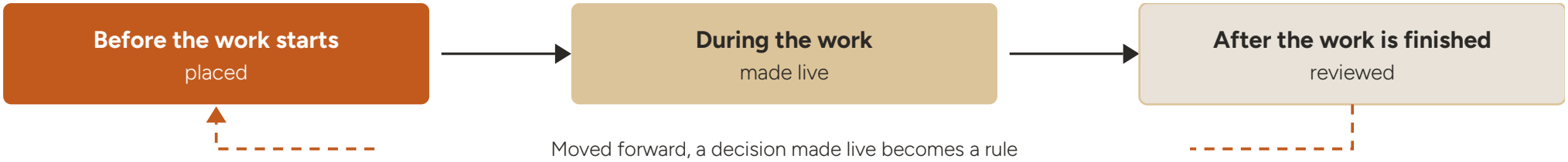
Symbols drive systems, and systems drive behaviour. The broken windows theory comes from Wilson and Kelling, "Broken Windows", The Atlantic, March 1982: minor disorder that is left unaddressed signals that no one is in control, and it invites serious crime. Giuliani targeted graffiti, fare evasion and aggressive panhandling, and did not tolerate any crime, however minor.

"The standard you walk past is the standard you accept." (a line made famous by Lt Gen David Morrison, Australian Army, 2013)

If Rudy Giuliani can turn a city of millions around by focusing on the small things, you can surely turn a shift and a pit around by focusing on the small things and enforcing the standards you expect!

Let's talk about decisions

Decisions live in three places: before the work starts, during the work, and after the work is finished.



The more decisions you can put before the work starts, the less we depend on variability and confusion during the work. For example, if we all know we don't walk diggers off benches or work areas, we park them, there can be no confusion.

Where	What lives there
Before the work: placed	Rules and golden rules, TARPs, contingencies with a name on them, work allocations, priorities, the minimum crew. Made once, written down, checked in Block 1.
During the work: made live	What no rule covers: a digger down with no contingency named, an allocation argued on the radio. It costs time and depends on who is on.
After the work: reviewed	The Shift Report and the final drive-around: which decisions were made live, what they cost, and which should become a rule.

A decision made under pressure depends on whoever is standing there. Most of what goes wrong on a shift is not new. It is the same few situations, decided again each time. Work actively with your Superintendent to find the decisions you are making again and again that lead to different ways of running a shift, and turn them into golden rules, so everybody operates with the same guard rails. The better you get at this as a team, the less chance there is of you coming back on roster and finding a pit that looks and operates completely differently from the way you set it up when you handed it over. Write each one down in Block 4 (chapter 3) and bring it to the final drive-around.

The test of a placed decision. Could a supervisor who has never met you run this shift the same way, because the decision is written down and not in your head?

Let's talk about assumptions

Your shift, your work area, your target and your plan are all based on assumptions. They can include all sorts of things: a fully staffed crew, all your machines available, the dig rate you can achieve, the number of loads per hour, the volume of material per truck, the distances they travel, good weather, dig faces big enough for the diggers to actually achieve their optimal swing and load rates, operators who know what they are doing, and on and on.

It is important that you know what assumptions are driving the targets you are given, and the design of the work areas the technical services team has designed for you, because if any one of those assumptions is not true, you may not be able to achieve your targets. For example, if a digger is working a bench too high for it, you will not get the optimal load rate from that machine. We cover optimal mining conditions in Part B of this program, but you must ask and understand what assumptions were used to create the targets you have been set.

These assumptions become your PADAL numbers (page 36).

Find these numbers, then check them against what you see

Assumption	The question to ask
Digging rate, by digger	What loads an hour, or bcm an hour, was priced for each digger, and are we at it?
Truck numbers and truck rate	How many trucks were priced on each digger, at what cycle time and payload?
Delay (SMU) allowance	What share of engine hours was allowed for delay on each type of machine?
Payload	What payload was assumed for each truck class, and does the spread of payloads sit inside the site's payload rule?
Ancillary factors	How many hours of support gear (dozers, graders, water carts) were allowed for every 1,000 bcm or tonnes moved?
Labour factor	How many people hours were priced for each machine engine hour?
Availability, by fleet	What availability was priced, and are we above or below it?
Haul distance and cycle time	What haul was planned, and has the dump or the dig moved it without the rate moving with it?

When you do this, use: Apply Tool 7, the assumptions I inherited. Redo it each quarter.

Too often, when volumes are not achieved, people add more stuff. More trucks, usually. This rarely works (unless it is an availability issue and they are buying redundancy: not your problem).

Think of a highway. If traffic is not flowing fast enough and not enough people are getting to work on time, adding more cars to "get more people through" makes it worse. Why would you do the same thing in a circuit? Fix the root cause of the issue. The shift and the circuit are yours! Own it.

You don't have to accept every crazy idea someone above you comes up with. But to push back, you had better know your facts, have your house in order and be on top of your game!

Trick for young players. Ask your digger drivers! They are usually very experienced operators and can tell you what they need to improve their load rates. Also ask them if they are happy with the set-up of the work area, and if safety or efficiency can be improved. You'll be surprised what you get back!

Know your numbers

Throughout this program we refer to the fact that you must know your numbers in loads and buckets per hour. This is because nobody can track 12,000 bcm. Broken down, the target becomes something you can check by standing still for two minutes, and something an operator can write on a card every block. Work your numbers out at the handover, from your own machines, this face and this material. Say them at the pre-start, and check against them every block. If the crew cannot say them, the target is not committed: it is only reported.

The number you commit is in loads. You report it in loads and buckets per hour. Buckets a minute and seconds a bucket are only for checking at the face. Write the number in one place: the top of the Shift Report, on day shift and night shift.

The questions you must ask

Step	The question
1 · Shift volume	What are we committed to this shift? One number, for this crew, this shift.
2 · Loads	How much does one truck carry on this circuit? The target payload, not the nameplate and not the best ever.
3 · Digging hours	How many digging hours have we really got? Shift hours less pre-start, travel, crib, fuel and shift change.
4 · Loads an hour	How many loads an hour does that mean, and how many per truck?
5 · Buckets a load	How many buckets fill a truck on this face, with this machine, in this material?
6 · Buckets a minute, seconds a bucket	What does that look like from the seat, at the face?

Say it in one sentence: "We need ___ loads this shift: that is ___ an hour, ___ per truck per hour, and a bucket every ___ seconds."

Calculate your numbers (before the pre-start)

The worked example. One example runs through the steps, with illustrative numbers. North Pit, day shift: one digger (EX1) and four trucks (T1 to T4). The target is 12,000 bcm at 60 bcm a load: 200 loads. First bucket 06:30, last bucket 16:30: 10 hours, and about 9 digging hours after crib and fuel. So about 22 loads an hour, about 5.5 per truck per hour. At 5 buckets a load that is about 111 buckets an hour: a bucket every 32 seconds or so.

Fill it in from your own machines, this face, this material. The worked example is only an example: replace every figure with yours.

Step	Calculation	Your figure	Worked example
1 · Shift volume	From the weekly plan, adjusted for today's machines: bcm or t		12,000 bcm
2 · Loads	Volume ÷ payload per load = loads		12,000 ÷ 60 = 200 loads
3 · Digging hours	Shift hours − pre-start − travel − crib − fuel − shift change = digging hours		12 − 3 (pre-start and travel, crib, fuel, shift change) = 9 h
4 · Loads an hour	Loads ÷ digging hours = loads an hour; ÷ trucks = per truck per hour		200 ÷ 9 = about 22 an hour; ÷ 4 trucks = about 5.5 per truck per hour
5 · Buckets a load	Buckets that fill a truck on this face, with this machine, in this material		5
6 · Buckets a minute, seconds a bucket	Loads an hour × buckets a load ÷ 60 = buckets a minute; 60 ÷ buckets a minute = seconds a bucket		22.2 × 5 ÷ 60 = about 1.85 a minute; 60 ÷ 1.85 = about 32 s

This crew's numbers: write them here, say them at the pre-start

Unit or circuit	Loads this shift	Loads an hour	Per truck an hour	Buckets a minute	Seconds a bucket	First bucket	Last bucket

Loads target, block by block: Block 1 ___ Block 2 ___ Block 3 ___ Block 4 ___ Shift ___

You will need to understand and master this skill, because we use it and refer to it across the whole program. Ask your Superintendent if you are unsure how to do this.



3. The five stepping stones that run a professional shift

There are five things a supervisor must do well to nail each shift. We cover those next.

Section 1 · The five stepping stones at a glance

Step 1 · The handover

Step 2 · The pre-start

Step 3 · Time in field: the four blocks

Step 4 · Time in office: the Shift Report

Step 5 · The handover to the next crew

The five stepping stones at a glance

The five stepping stones a supervisor must nail every shift are:

Step	The stone	Block	Where
Step 1	The handover	Before Block 1	The office
Step 2	The pre-start	Before Block 1	The pre-start room
Step 3	Time in field: the four blocks	Blocks 1 to 4	The field
Step 4	Time in office (in Block 4): the Shift Report and other admin	Inside Block 4	The office, only as long as you need
Step 5	Handover (in Block 4): the final drive-around and the handover to the next crew	Inside Block 4	The field, then the office

You cross the five stepping stones in order, every shift. Miss one and the gap carries into the next shift.

Steps 4 and 5 do not come after the four blocks: they happen inside Block 4. Then the next crew goes back into the field, and the cycle starts again.

For each of these steps there is a 'what good looks like' section and a coaching card you can use to track your performance. You will hand these coaching cards in, with your own Yes/No evaluations, at the end of each of the five stepping stones and the four blocks of the time in field section, which we cover below.

The cards are your cheat sheets. Each coaching card is a stand-alone tool. Take it from the Apply tools pack and take the card for each step with you: you do not need to carry this book on shift. Steps 1 to 4 each have a card; Step 5 uses the handover card. Card lines continue across the cards.

Your Apply tools (issued with this handbook): Tool 1, work out your number · Tool 2, the operator's count card · Tool 3, the operator scorecard · Tool 4, the handover sheet and cards · Tool 5, the four block cards · Tool 6, the Shift Report · Tool 7, the assumptions I inherited · Tool 8, the six behaviours self-evaluation · Tool 9, the fault-tree tool.

Let's now look at each stepping stone in sequence.

Step 1 The handover

Key question: Do I understand the real state of the pit, the plan I will run, my numbers, and what I must set up for the next shift?

The handover is where you take over the shift. It is held in the office, with all the supervisors and Superintendents there. You get the off-going supervisor's account of the real state of the pit and the plan you will run. The pit plan and layout are checked in the office against the mud maps or pit maps. It is also where you are told what you need to do to leave the next shift ready to run. Done well, you can state the plan and the pit without notes. You leave knowing your numbers in loads and buckets per hour (see page 14), ready to give them to the crew. The handover does not commit the number: that waits until you have seen the ground in Block 1 (Step 3, time in field, page 24). Next, you take the plan and the numbers to the crew at the pre-start.

Outcome: Before the pre-start, you hold the real state of the pit, the plan checked across processes, your numbers, the focus areas and the contingencies for the shift, and what you must leave ready for the next shift.

- **Owner:** the off-going supervisor
- **With:** you, the oncoming supervisor; all supervisors and Superintendents on shift. You fill in the card and self-evaluate; the Superintendent coaches you on it
- **When:** 05:00, 30 minutes, in the office. Times are indicative. The off-going supervisor inspects the next shift's areas in Block 4 of their own shift, before the handover.

Decisions the handover must make (tick each)

#	Decision	Done
1	The priorities for the day	☐
2	The contingencies for the day, to mitigate the identified risks	☐
3	The allocation of gear and people	☐
4	What you must leave ready for the next shift	☐

Your handover agenda and what to bring

Agenda (30 minutes)

Min	Item	Lead	Output
7	Last shift in PAVA, from the report: Plan, Actual, Variance/Impact, Actions taken	Off-going supervisor	The real state of the shift
5	The pit plan and layout, checked against the mud map or pit map	Off-going supervisor	Hazards, dig position, floor and roads
5	What the last shift left set up for you, and what you must set up for the next	Off-going supervisor	Written, each with a name
5	The plan across processes: staff, gear and targets	Oncoming supervisor	Changes agreed with the Superintendent
3	The contingencies, TARPs and golden rules in force	Oncoming supervisor	Each contingency still right, with a name
3	The focus or priority areas	Oncoming supervisor	Written on the plan
2	The things that were escalated and still need to be resolved	Oncoming supervisor	Each issue with a name

Bring to the handover

What	Who
Last shift's report, in PAVA (Plan, Actual, Variance/Impact, Actions taken)	Off-going supervisor
The plan for this shift	Off-going supervisor
What the off-going supervisor found inspecting the next shift's areas in Block 4	Off-going supervisor
The mud map, pit map or mine plan for your circuit	Mine Planning, the Superintendent
The targets for the shift	The Superintendent
The plans for drill, blast, ROM and maintenance	Mine Planning, the Superintendent
The safety rules, TARPs, golden rules and contingencies in force	The Superintendent (the commitments register)
The staff on site: who is here, who is away, who is new	Off-going supervisor, from the roster
The available gear: what is running, what is down or booked for service	Off-going supervisor, from maintenance
Expected shutdowns, outages, weather, water and closed areas	The Superintendent

Rule: a job to set up, or a risk, without a name is not handed over. Anything Block 1 shows did not hold goes to the Drive-around.

Coaching card: the handover

Yes / No

Stand-alone tool: Apply Tool 4. Use it at both handovers: this one, and the handover to the next crew in Step 5. You fill in this card and mark each line Yes or No yourself, as you self-evaluate. Your Superintendent coaches you on it.

#	What good looks like	Y/N
1	Started and finished on time, with all supervisors and Superintendents there; meeting behaviours kept.	
2	The plan for this shift was ready before 05:00 and used in the handover.	
3	Last shift taken over in PAVA from the report: dig position, rates, hazards, what was escalated, work not finished.	
4	Pit plan and layout checked in the office against the mud map or pit map, area by area; understanding checked.	
5	Targets set for the staff and gear really there this shift, not the budget or an average.	
6	The focus or priority areas for the shift agreed.	
7	The contingencies in force reviewed, with the TARPs and golden rules; each contingency has a name.	
8	The next 24 hours checked across drill, blast, load and haul, ROM and maintenance: services, supplies, gear and order of work.	
9	Each risk to the plan either planned around or escalated, with a name on it.	
10	Told what this shift must set up for the next. Checked what the last shift left set up. Each item has a name.	
11	What did not fit was challenged; changes agreed with the Superintendent before pre-start.	
12	Key actions written on the plan as tasks for the crew, ready for pre-start.	
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	

Actions required from any No: ___ Owner: ___ By: ___

When you do this, use: Apply Tool 1, work out your number, and Tool 4, the handover sheet.

Step 2 The pre-start

Key question: Does every operator leave knowing their machine, area, target and fallback?

The pre-start gets the crew to their gear knowing the job. You run it; the Superintendent may watch but does not brief. Talk slowly and clearly, and check understanding by asking the operators, by name. For example: "Tony, what are our contingencies for the day?" "Jack, what is our safety focus for the day?" Every operator must be able to say when first bucket and last bucket are, the targets for the day, the contingencies and the priority areas. Ask the crew what the risks are to making the numbers today, and what the safety focus is. And ask the question every shift: "What did we get away with yesterday?" Then Block 1 starts. The pre-start does not re-make what was decided at the handover.

Operator scorecards: your most useful tool. Keep it on the wall and up to date, in the room where everyone can see it. It shows each operator's trend over the roster, not a comparison with the last shift. At least once a roster, name and congratulate the top performers in front of the crew. Never call out poor performers in front of the crew.

Outcome: A briefed crew out of the room by 06:15 and straight to their gear, each operator knowing their task.

- **Owner:** you, the supervisor
- **With:** the crew and leading hands; the Superintendent watches. You fill in the card and self-evaluate; the Superintendent coaches you on it
- **When:** 06:00, 15 minutes. Times are indicative.

Decisions the pre-start must make (tick each)

#	Decision	Done
1	Fit to start, area by area: people, machines and conditions, or that work does not start	☐
2	Anything raised by the crew that changes the plan, decided on the spot	☐

What to bring and your pre-start agenda

Bring to the pre-start

What	Who
The plan board: each operator's machine, area and task	You
Your numbers: the targets in loads and buckets; first and last bucket times	You, from the handover
The contingencies, in if/then words, and the priority areas	Handover
The safety focus for the day	You
The operator scorecard, up to date: the trend over the roster	You

Agenda (15 minutes)

Min	Item	Lead	Output
3	Safety: today's hazards, areas and safety focus; what did we get away with yesterday?	Supervisor	Hazards named, with their controls
3	The targets, in loads and buckets; first and last bucket	Supervisor	The crew can say them back
2	Last shift, area by area, and the operator scorecard	Supervisor	At least once a roster, the top performers named
5	The plan, per operator; the contingencies and priority areas	Supervisor	Every task given
2	Said back, by name: first and last bucket, the targets, the contingencies, the priority areas, the risks to the numbers and the safety focus	Supervisor	The crew to their gear

Rule: an operator who cannot say first and last bucket, the targets, the contingencies and the priority areas is a briefing failure. Fix it before they leave the room.

Coaching card: the pre-start

Yes / No

Stand-alone tool: the pre-start card in Apply Tool 4. You fill in this card and mark each line Yes or No yourself, as you self-evaluate. Your Superintendent coaches you on it.

#	What good looks like	Y/N
13	Started and finished on time, with the whole crew there; meeting behaviours kept.	
14	Run from the plan made at the handover, checked by the supervisor before the meeting.	
15	Safety first: today's hazards, areas and safety focus, in more detail when something has changed or the crew is new.	
16	Last shift's results given briefly, area by area.	
17	Each area and its key work described; every operator given a machine, an area and a task.	
18	The shift target said to the crew in loads and buckets, with first and last bucket times.	
19	Operator scorecard up to date and on the wall where all can see it; at least once a roster, the top performers named and congratulated, on the roster's trend.	
20	Spoken slowly and clearly, with authority and energy.	
21	Understanding checked by name: operators can say first and last bucket, the targets, the contingencies, the priority areas and the risks to the numbers; a safe and productive shift expected.	
22	Crew out of the room within 30 seconds of the end, straight to their gear.	
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	

Actions required from any No: ___ Owner: ___ By: ___

When you do this, use: Apply Tool 1, work out your number, Tool 3, the operator scorecard, and the pre-start card in Tool 4.

Step 3 Time in field: the four blocks

Key question: Am I running an effective shift? Am I going to hit my numbers this shift, and if not, what can I change during the shift to save it?

Instead of running your shift as 'time in field' with no real structure, this program breaks it into four blocks of two to three hours. Each block has one clear focus. Blocks 1 to 3 end with a meeting with your Superintendent: the Drive-around, the touch-base and the check-in. You also spend time in the cab with your Superintendent: at the end of Block 1, once your day is under way, and in Block 4, so you are both aligned on today and ready for the next shift.

The blocks give your day structure and control. You catch problems inside the shift, while there is still time to fix them, instead of losing a whole shift and talking about what went wrong the next day. Because everyone knows who is doing what in each block, the constant calls, the unnecessary meetings and being pulled from pillar to post stop. Your blocks are linked to your Superintendent's day, so your day is yours to run again.

Outcome: A well-structured, disciplined and orderly shift that runs to plan, which gives you back the time you need to do your job well.

- **Owner:** you, the supervisor
- **With:** the crew; the Superintendent at the Drive-around, the touch-base, the check-in and the final drive-around
- **When:** 06:15 to the end of the shift. Times are indicative.

Block 1 done well. First bucket on time; all operators to their gear on time; go-lines operating efficiently; fuel plan in place and kept to; all work areas inspected and set up for safety. You get clear on the realities of the shift, verify that the handover you got is correct, and confirm whether the shift target is achievable with the crew actually on shift, the gear available and the pit conditions as expected.

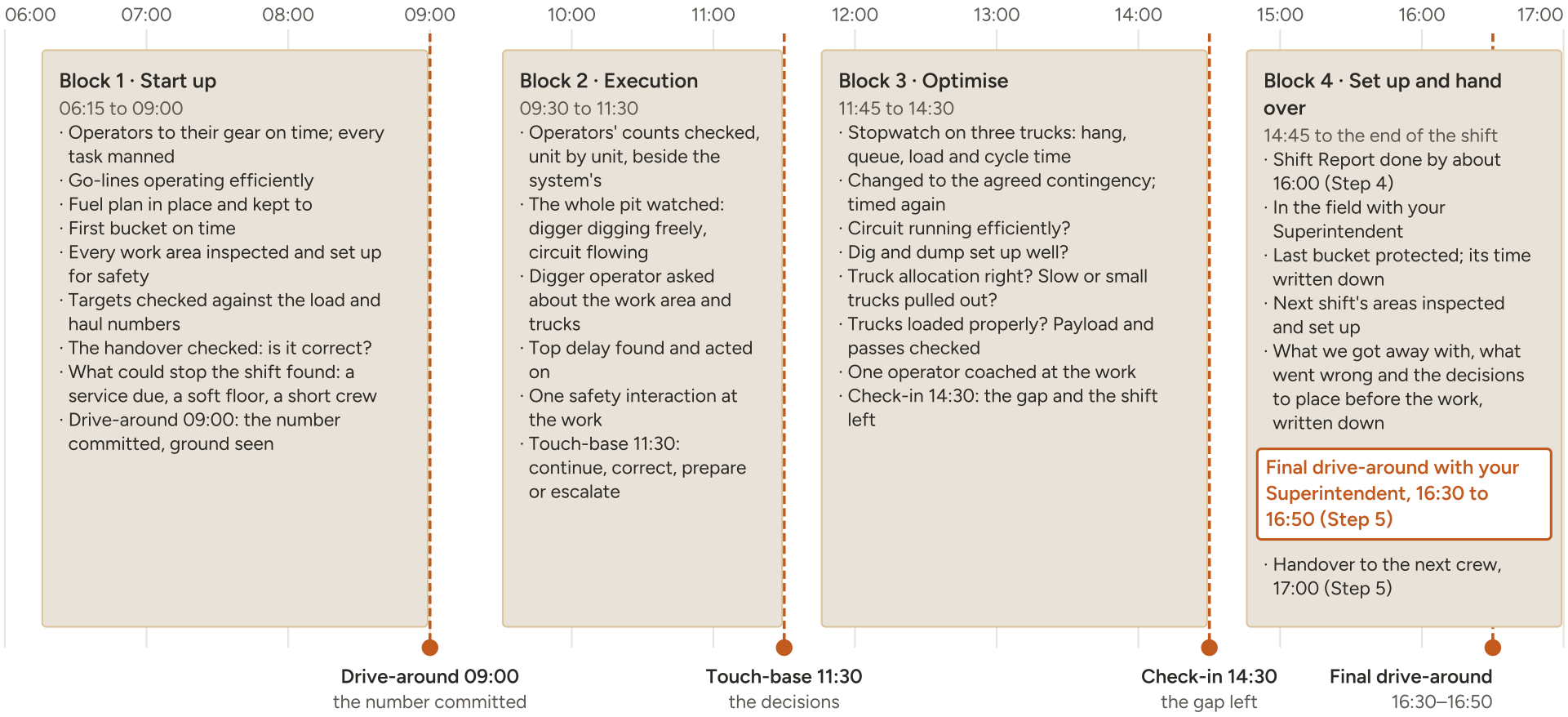
Block 1: the Drive-around. Bring what you saw, not what the plan said. If the number you think you can achieve is less than the target, be clear why: what on the ground is the reason, and which assumption has moved. Then say exactly what you need fixed that you cannot fix yourself. Write the number at the top of the Shift Report before Block 2 starts.

Block 2: ask the digger operator. Get on the radio and ask the digger operator whether their work area and truck allocation are set up the way they need, or whether something needs to change (see "Trick for young players" in chapter 2). Your safety interaction is a short, two-way talk with an operator about the job in front of them.

Block 3: look for better work. If your number is at risk, change to the agreed contingency if one fits, say what you expect the change to do, and time it again. Do not stop at one thing. Your job is the quality of the work, not just getting tasks done: keep define, allocate, coordinate and verify in mind. Chapter 4, Diagnostics, shows where to look and what each stopwatch time tells you. Block 4 follows on pages 25 to 26.

Your four blocks, and what you check in each

The four blocks are fixed, and so are the touch-base and the check-in. Timing changes need your Superintendent's approval. Each bar below holds everything you check in that block.



The four blocks across the shift, with everything you check in each. Orange markers: where you meet your Superintendent. Times are indicative.

3. The five stepping stones that run a professional shift

Block 4: finish strong. Runners do not slow down before the finish line. Your team shouldn't either. Leave floors cleaned, gear where it is needed, and nothing broken for the next crew to find. There is no check-in in Block 4. What is critical: the next shift is set up and the last bucket is protected.

Night shift. The same blocks, with no Drive-around, no check-ins and no final drive-around, because there are rarely Superintendents on duty on night shift. If there are, revert to the day-shift four blocks. The last block of night shift is still about protecting the last bucket, making sure the next shift is ready to run, and creating a robust handover. Write the number you commit at the top of the Shift Report, as on day shift. Escalate by radio or phone when recovery is beyond the shift.

How your blocks link to your Superintendent's day

Your four blocks are linked to your Superintendent's four blocks, on the same clock (side by side, page 27).

When do you spend time with your Superintendent?

Your Superintendent does their own drive-around in their Block 1, after pre-start. At the end of your Block 1 you two have a combined drive-around, so you can tell them what you see and they can tell you what they see. This is where you get to say what number you can achieve, and whether the handover you got and the assumptions made for that shift are accurate and correct. Don't skimp on this, and don't get dragged into their car for half your block! Protect your block.

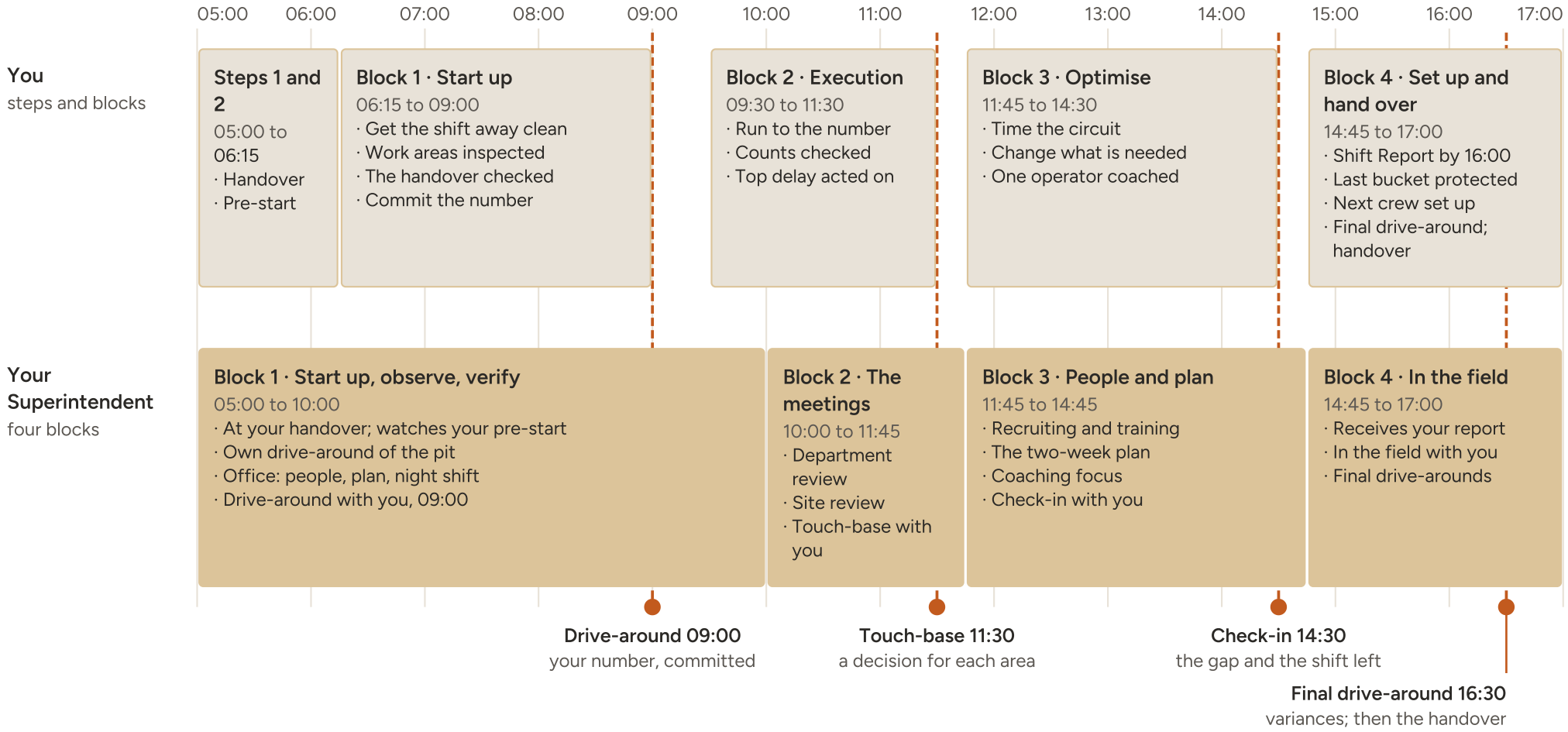
You will need to touch base with them midday, preferably at the lookout, and you will need to know your performance numbers for the first part of the day: how many loads, how many buckets, are you going to hit the number **you** committed to? If not, why not, and what can you do to improve and save the shift? If anything new has popped up that you need to escalate or make them aware of, now is your chance.

In Block 4, once your write-up is done, you and your Superintendent are in the field together until the last bucket, then do the final drive-around.

You are not at the Daily Department Review, the Daily Site Review or the weekly and monthly reviews. You are in the pit. With several supervisors, the Superintendent sees each of you in turn: Drive-arounds from 09:00, touch-bases from 11:30, check-ins from 14:30, final drive-arounds from 16:30. Your block runs on until your turn, and your next block starts after it.

Your blocks and your Superintendent's blocks, side by side

Your day and your Superintendent's day on one clock: your blocks on top, theirs underneath. The orange lines are where you meet and something passes between you.



One clock. Orange lines: where you meet your Superintendent and something passes between you. Times are indicative.

Coaching card: the four blocks

Yes / No

Stand-alone tool: the four block cards in Apply Tool 5. You fill in this card and mark each line Yes or No yourself at the end of its block, as you self-evaluate. Your Superintendent coaches you on it. A No is the next thing to fix: write the action on the card.

#	What good looks like	Y/N
Block 1 · Start up		
23	First bucket on time; all operators to their gear on time.	
24	Go-lines operating efficiently; fuel plan in place and kept to; all work areas inspected and set up for safety.	
25	The handover you got verified as correct.	
26	The shift target confirmed as achievable with the crew actually on shift, the gear available and the pit conditions as expected.	
Block 2 · Execution		
27	Loads, buckets, payloads from the operators' counts, beside the system's; differences settled.	
28	Top delay acted on; one safety interaction at the work.	
29	Touch-base 11:30, in numbers: continue, correct, prepare or escalate.	
Block 3 · Optimise		
30	Stopwatch on hang, queue, load and cycle time; what was needed changed and measured.	
31	One operator coached at the work.	
32	Check-in 14:30: the gap, and the shift left.	
Block 4 · Set up and hand over		
33	Last bucket protected.	
34	Next shift's areas inspected and set up.	
35	What we got away with, what went wrong and the decisions to make before the work, written down.	
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	

Actions required from any No: ___ Owner: ___ By: ___

When you do this, use: Apply Tool 2, the operator's count card, and Tool 5, the four block cards.

Step 4 Time in office (in Block 4): the Shift Report

Key question: How did the shift go against the number I committed, and why?

The time in office at the start of Block 4 is for you to use only if you **must** do something in the office. Keep office time as short as possible: your job is in the pit, not behind a desk. You can use it to complete your shift report if you need to do it in the office, but be careful not to get sucked into unnecessary office meetings. Be clear about what is and is not your job: the fewer meetings you go to, the better.

The Shift Report is the honest account of the shift against the number you committed. You write it through the blocks, from the operators' counts, and finish it in Block 4. Done well, the next supervisor could run their handover from it alone. It goes to the Superintendent. You explain it at the final drive-around and hand over from it in Step 5.

Outcome: A report done by about 16:00, block by block, from the operators' counts: the gap in one line, every variance explained, and a decision on every variance. The final numbers can be added when the last block is done.

- **Owner:** you, the supervisor
- **With:** the operators, whose counts it is built from
- **When:** through the shift, block by block; finished in Block 4, done by about 16:00. Times are indicative. Your site can change them.

What a Shift Report must capture. Your shift in PAVA (Plan, Actual, Variance/Impact, Actions taken), in loads and buckets per hour, block by block and by digger, with the delays, what you got away with, the jobs not finished and the risks the next shift must plan in.

Coaching card: the Shift Report

Yes / No

Stand-alone tool: the report card in Apply Tool 6. You fill in this card and mark each line Yes or No yourself, as you self-evaluate. Your Superintendent coaches you on it.

#	What good looks like	Y/N
36	Written on the Shift Report sheet through the blocks, from the operators' counts, not from memory.	
37	Plan against actual, in PAVA, in loads and buckets per hour, block by block and at the end, with any statutory reports.	
38	Jobs not finished this shift are listed.	
39	It shows what was done, and when, each time the shift went off plan.	
40	The gap in one line; each variance explained (its impact, the action taken and whether it worked) or accepted in writing.	
41	What could have foreseen or avoided each variance, and whether it reaches the next shift, written down.	
42	Risks to the next shift written down, with the jobs it must plan in.	
43	Done by about 16:00, before the push to the last bucket.	
44	Explained at the final drive-around and used at the end handover with the oncoming supervisor.	
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	

Actions required from any No: ___ Owner: ___ By: ___

When you do this, use: Apply Tool 6, the Shift Report.

The Shift Report template (page 1 of 2)

Write it through the blocks; finish it in Block 4. The blank is Apply Tool 6. Every number is in loads and buckets per hour, in PAVA: Plan, Actual, Variance/Impact, Actions taken.

Supervisor: ___ · Crew: ___ · Superintendent: ___ · Date: ___ · Shift: Day / Night · Pit and areas: ___

The shift target: ___ loads · The number I committed at the Drive-around (written before Block 2): ___ loads

Production by digger

Digger and area	Material	Loads plan	Loads actual	Loads an hour (plan / actual)	Buckets an hour (plan / actual)	Payload (plan / actual)	Digging hours (plan / actual)	Variance and impact	Action taken

Block	Loads plan	Loads actual (my figure)	The system	Variance and impact	Action taken
1 · Start up					
2 · Execution					
3 · Optimise					
4 · Set up and hand over					
Shift					

Measure	Plan	Actual	Variance and impact	Action taken
First bucket				
Last bucket				
Fuel plan				
Cycle time				
Trucks on the circuit				

The Shift Report template (page 2 of 2)

Safety and the pit

Item	What I found	Action taken
Work areas inspected		
Safety interactions held		
Hazards found, and their controls		
What we got away with		
Incidents and near misses reported		

Delay	Unit	Minutes	Cause	Action taken

Support work and tasks	Planned	Done	If not, why
Dozers			
Graders and water carts			
Scheduled tasks and compliance			
Statutory reports			

- The gap, in one line (the number and the cause): ___
- Each variance: what could have foreseen or avoided it, and does it reach the next shift: ___
- What went wrong, and the decisions to make before the work: ___
- Jobs not finished, and what the next crew must finish: ___

Signed (supervisor): ___ Received (Superintendent): ___

- Risks to the next shift, and the jobs it must plan in: ___
- Escalated, and still to be resolved: ___
- Emerging issues, or changes to the plan for the next shift: ___
- Key actions I need from the next shift: ___

Step 5 The handover to the next crew (in Block 4)

Key question: Is the next crew ready to start well, and do they know what I need from them when I come back?

Step 5 is the second handover of the shift, in Block 4. It comes after your write-up (done by about 16:00) and your time in the field with your Superintendent until the last bucket, at about 16:30. During your final drive-around with your Superintendent, you explain the variances against the plan and their impact: what actions you took and whether they worked, what could have been done to foresee or avoid them, and whether they will reach the next shift. Then you hand over to the next crew, and they go back into the field.

Outcome: A completed handover for the next crew, with the key actions you need from the next shift so your next shift starts well when you return.

- **Owner:** you, the supervisor
- **With:** your Superintendent, at the final drive-around; the oncoming supervisor, with all supervisors and Superintendents on shift, at the handover
- **When:** after the last bucket: the final drive-around, 16:30 to 16:50 (with two supervisors, 15 minutes each, at 16:30 and 16:45; with three, 10 minutes each, at 16:30, 16:40 and 16:50; all end by 17:00); the handover at shift change, 17:00. Times are indicative. Your site can change them. Night shift: you explain the variances at the start of the morning handover, with the day Superintendent.

Decisions the handover to the next crew must make (tick each)

#	Decision	Done
1	The things that need to be escalated are clearly defined and agreed	☐
2	The key actions you need from the next shift	☐

What to bring to the final drive-around and the handover

What	Who
The four block cards	You
The Shift Report	You
The handover sheet, filled in for the next crew	You
The actions page, and the last coaching point	The Superintendent

Agenda and coaching card. This is a handover. Run it to the handover agenda in Step 1, and mark it on the handover card in Step 1 (lines 1 to 12). There is no separate agenda or card.

Rule: you explain the variances; your Superintendent checks them, never the other way round. A cause seen for the third time leaves as a system finding for the week, with the evidence. Your Superintendent may leave you one coaching point; it is checked at the next final drive-around.

When you do this, use: Apply Tool 6, the Shift Report and its actions page, and Tool 4, the handover sheet and card.



4. Diagnostics

When the number is at risk, find out why, and fix the cause, not the symptom.

Section 1 · Reading the mine

Section 2 · How your performance is measured: PADAL

Section 3 · Productivity levers

Section 4 · Finding the cause

Section 5 · The fault trees

Reading the mine: your part

Your circuit is one part of a chain. Drill, blast, load and haul, ROM and maintenance each hand work to the next. Your circuit loads and hauls what drill and blast made ready. It delivers to the ROM or the dump, and runs on the units maintenance hands back. A delay in any one of them shows up in your loads. That is why the handover checks the plan across all five.

Know your contract. How the contract pays decides what counts. If it is paid on rates and volume, the money is in the volume moved: the machines are charged for every hour the engine runs, and an engine hour that moves no volume is paid for and earns nothing. If it is paid on SMU hours only, it is usually just a staff rate and an hours rate. Part B covers this in more detail, but know your site's contract now: the PADAL measures your shift is judged on may change with it (see "How your performance is measured: PADAL", below). Ask your Superintendent which of the five your site reports most. On any contract, a busy-looking pit is not the test. Loads an hour against the number is.

Your numbers. The target, broken down into loads an hour and buckets a minute, is worked out at the handover and said at the pre-start. When the number is at risk, the rest of this chapter shows how to find out why. Loads and buckets per hour are worked out in chapter 2: "Know your numbers" (page 14) and "Calculate your numbers" (page 15).

How your performance is measured: PADAL

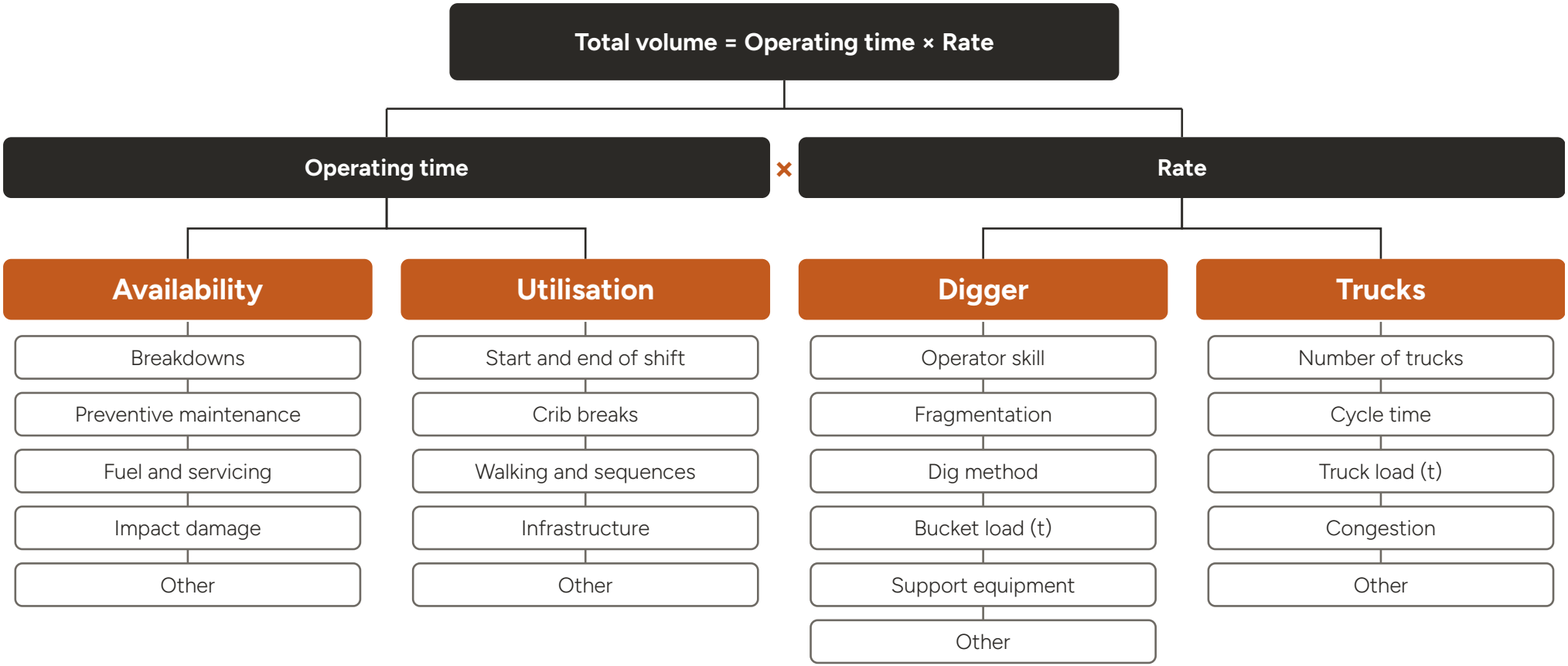
Your shift is measured on five numbers, called PADAL. Every one of them moves on your shift. The loads and buckets your operators count are how you move them. PAVA is how you report; PADAL is what you are measured on. When you commit under target at the Drive-around, say which PADAL number moved and why: "D is up because we waited 40 minutes for the blast to clear" is better than "the blast crew let us down".

Letter	The number	In plain words	Where your shift moves it
P	Primary productivity	How much is moved for each hour the digger is loading. Higher is better.	First bucket on time, last bucket protected, hang time, buckets a minute.
A	Annualised work hours	How many hours the gear works, scaled up to a full year. Higher is better.	Every hour of delay, queue or waiting takes an hour off it.
D	Delay SMU percentage	The share of engine hours lost to delay instead of work. Lower is better.	Waiting for trucks, fuel or the blast: coded honestly in the Shift Report.
A	Ancillary factors	Hours of support gear (dozers, graders, water carts) for every 1,000 bcm or tonnes moved. Lower is better.	Support gear on task and to plan, re-tasked only by an agreed contingency.
L	Labour factor	People hours for each machine engine hour. Lower is better.	Seats manned against machines; a spare operator used, not parked; no truck manned that the circuit does not need.

Productivity levers

Key question: Which of these levers could or should be pulled to increase productivity around your operation?

In summary, there are only two levers that you as a supervisor can pull to improve productivity: operating time and rate. They are below. This is the first stop when you reflect on the shift: start here.



Reading the levers

Operating time is availability × utilisation. Rate is what the digger and the trucks do with the time they get. Two more causes sit under the rate on shift: the floor the digger works on, and the trucks' spot and queue time at the face.

Every delay you code, and every rate you lose, lands on one of these levers. Name the lever before you name the fix. "We need to lift productivity" is a wish. "We lose 40 minutes a shift to start-of-shift utilisation, and here are the three causes" is a decision waiting to be made. The rest of this chapter, and the fault trees at the end of it, help you find the cause behind the lever.

Finding the cause

Your job in the field is the quality of the work, not just getting tasks done. A busy pit can still be losing the number. Use this chapter in any block, and above all in Block 3: first name the productivity lever, then look at the circuit, the dig and the dump, the truck allocation and the loading.

Which part of management failed? Every fix falls under define, allocate, coordinate or verify (the definition of management, page 9). When something goes wrong, ask which one failed.

The fault trees. The fault trees support the productivity levers: they take a lever down to its cause. Fault tree 1 sits under the rate (the digger and the trucks); fault tree 2 sits under operating time (availability and utilisation). When a loss keeps coming back, walk the fault tree. Start at the top and work down. Do not stop at the branch: the branch is the symptom. Keep going until you reach a box you can change, then check the foundation bar, because most branches end there.

How to use it: take the single biggest loss of the period and walk the tree out loud with your crew or your Superintendent, naming the box at each level. Where two people name different boxes, you have found a data problem, not a disagreement. Write down the lowest box you reach as the cause. If you land on the foundation bar, say so plainly; do not call it an equipment fault.

Underneath every branch: the foundation bar. Management · Standards and procedures · Roles and decision rights · Competence and training · Supervision in the field · Culture.

Checking the circuit, the dig and the dump

Circuit efficiency. Stand where you can see the whole pit. Then put a stopwatch on three trucks, not one:

- **Hang time:** the digger waiting, bucket full, for a truck to spot.
- **Queue time:** a truck waiting at the face for the one in front.
- **Load time:** spot plus load, truck stopped to last bucket.
- **Cycle time:** face to dump to face.

Compare them with the target cycle for this haul. Hang and queue together tell you whether the circuit is short of trucks or has too many. Load time tells you about the face and the operator. Cycle time tells you about the road. Industry studies suggest a 10% rise in waiting time costs a few per cent of actual productivity (indicative).

The dig set-up: the loading face. Good looks like: the digger loads without waiting, trucks arrive already in position, and the payload sits in its band. The circuit is usually lost in the last 50 metres. Loading from both sides (double-side loading) removes truck exchange and is often credited with a gain of a few per cent over loading from one side (indicative). Set your bucket rate from your own machines. Check:

- bucket-to-bucket time: the gap between one bucket landing and the next;
- spot time: how long a truck takes to get into position;
- the floor, the toe and the face position, and bench height against the machine.

The dump set-up: the dump and tip head. Good looks like: trucks tip without hesitation, the edge is to standard every round, and the order of tipping matches the plan. Check:

- the tip-head edge and windrow: to standard, and lit at night;
- the dump level and design width: survey confirming the room left;
- the dozer on task: pushing to design, not idling;
- the dump sequence against the plan: cells used in the order planned.

Haul roads and intersections. Good looks like: trucks hold a steady speed without hunting for gears, the surface is kept up before it breaks down, and the grader works to a plan. Haulage can be more than half of the operating cost in a deep pit. Industry studies suggest that managing road condition and grade together cuts cycle time and fuel, and lifts each truck's output (indicative; it varies by site). Check the surface (corrugations, potholes, soft spots, loose rock), the width, grade and camber, the intersections (sight lines, signs, who has priority), and where trucks change gear: a grade that makes them hunt is costing you.

Truck allocation. Match the trucks to the circuit. Too many trucks slow a circuit and burn fuel: queue time goes up. Too few leave the digger hanging. Sometimes sending an operator to crib costs less than manning a truck you do not need.

Loading: are your trucks loaded properly? How do you know? Check the payload against the target for each truck: under-loading wastes the circuit, and over-loading wastes the truck. Count the passes (buckets) a load on this face. If it takes more passes than the target, find out why: bucket fill, fragmentation or the operator. Where the site has payload readings, put them beside the operators' counts. Ask your Superintendent what payload your circuit was priced on (chapter 2, "Let's talk about assumptions").

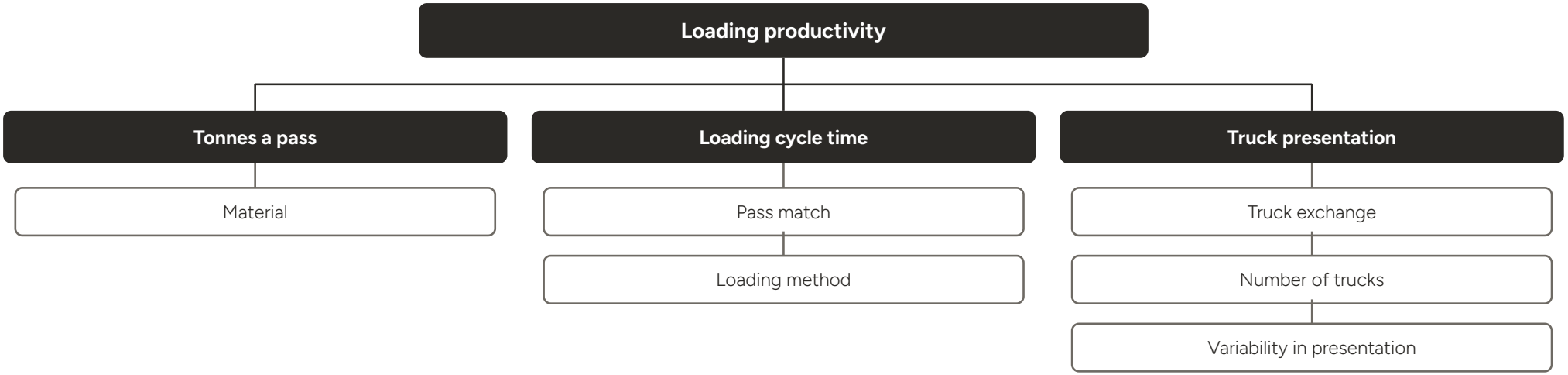
The enablers. None of these moves a tonne on its own. All of them decide whether the circuit can.

- **Water and pumps:** the next face is dry before the plan reaches it, and the wet-weather fallback is named and known. Ask every Block 2 whether dewatering is keeping up.
- **Drill and blast:** enough broken rock in front of the diggers that a hard band or a late blast does not stop digging. Ask every Block 2 whether drill and blast is on track for what you need next.
- **Support gear, fuel and the go-line:** dozers, graders and water carts where the plan needs them, before it needs them; fuel in its window; no unit parked without somebody knowing why. This one is yours: own it.

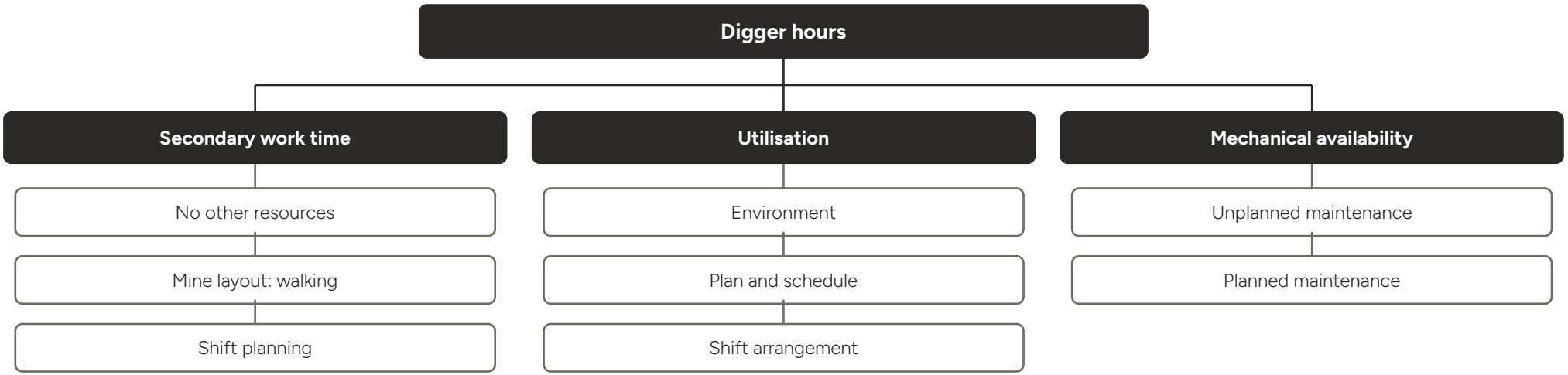
The fault trees

The full trees, every branch to its root, are the A4 fault-tree tool (Apply Tool 9).

Fault tree 1: loading productivity



Fault tree 2: digger hours (primary work time)



Fault tree 1, branch by branch

Branch	Second level	Third level, and what sits under it
Tonnes a pass	Material	How loose the muckpile is: design · drilling · shotfiring
Loading cycle time	Pass match (number of passes)	Bucket: maker's standards · other buckets available. Truck tray: payload limits · other trays available
Loading cycle time	Loading method	Planning (week and beyond): design · geology · objectives. Work area: shift planning · standards · operator competence. Material: several types
Truck presentation	Truck exchange	Work area: planning · competence · supervision. Competence: training · supervision
Truck presentation	Number of trucks	Shift planning: week planning · supervision. Availability: planned · unplanned
Truck presentation	Variability in presentation	Number of trucks: supervision · mechanical availability · operator. Different destinations: planning · material types · supervision. Haul road interference: planning · supervision. Competence: training · supervision

Fault tree 2, branch by branch

Branch	Second level	Third level, and what sits under it
Secondary work time	No other resources	Shift planning: supervision · weekly planning
Secondary work time	Mine layout (walking)	Design · standards and procedures
Secondary work time	Shift planning	Supervision
Utilisation	Environment	Wet. Wet recovery: haul road building · supervision. Fog and lightning: standards and procedures · geography
Utilisation	Plan and schedule	Go straight to the foundation bar. Was the plan made before the work, and was it followed?
Utilisation	Shift arrangement	Blast delays: shift planning · week planning. Crib delays: supervision · standards · data to track
Mechanical availability	Unplanned maintenance	Accident damage: work area · how production works · how maintenance works · environment. Routine maintenance missed: defects · machine housekeeping · short of skilled labour. Stores and supply: warehouse space · critical spares planning. Labour productivity: skills
Mechanical availability	Planned maintenance	Maintenance planning: standards · sequencing. Labour productivity: skills · supervision · taken off to unplanned work. Component life: target lives · operating conditions. Age of equipment: policy · replacement capital



5. Terms and academic models

Section 1 · Terms, A to N

Section 2 · Terms, O to Z

Section 3 · The evidence behind the model

Terms, A to N

Actions page: the page of the Shift Report where the actions from the final drive-around are written, each with an owner and a date.

Availability: the share of time the gear is fit to work: not broken down, and not in for service or fuel.

bcm: bank cubic metres: rock measured in the ground, before it is blasted.

Block card: the coaching card for one block: Yes or No lines, marked at the end of the block, with the block's numbers.

Block: one of the four parts of the shift in the field, two to three hours each, with its own card. Blocks 1 to 3 end with a meeting with your Superintendent.

Check-in: 15 minutes with the Superintendent, face to face, at 14:30 on day shift, in numbers. The 11:30 one is the touch-base.

Circuit: one digger and the trucks that haul from it, from the face to the dump and back.

Coaching card: Yes or No lines that say what good looks like. A No is the next thing to coach, not a mark against you. Each card is a stand-alone tool you can take on shift.

Commitments register: the written list of rules and contingencies in force, carried from shift to shift.

Contingency: an if/then fallback decided before the work, with a name on it.

Continue, correct, prepare, escalate: the four decisions each block and check-in ends in. Continue: on track. Correct: off the number, fixed inside this shift. Prepare: set up for the next shift. Escalate: beyond the shift's authority.

Crib: the meal break.

Cycle time: a truck's round trip: queue, spot and load, haul, dump, and back to the face.

Daily Department Review: the Superintendent's 10:00 meeting with the department's leads; what the supervisors cannot fix goes here, and the fix comes back to you in the pit.

Daily Site Review: the 10:30 meeting where the departments settle what one department cannot settle alone.

Double-side loading: the digger loads trucks on both sides, so one truck spots while the other is being loaded.

Drive-around: the Superintendent driving the pit with you after Block 1, where you commit the number.

Face: where the digger is digging.

Fault tree: a diagram that breaks a loss into its causes, branch by branch, down to a cause you can change.

Final drive-around: the last meeting with the Superintendent, in Block 4, where you explain the shift's variances (the debrief).

Four core components of management: define the work, allocate it, coordinate it and verify it.

Fragmentation: how well the blast broke the rock; good fragmentation digs faster.

Go-line: where the gear is parked between shifts, ready to start.

Golden rules (guard rails): the site's standing rules, never decided again by whoever is on duty. They grow as decisions made live become rules.

Handover sheet: the sheet in Apply Tool 4 the off-going supervisor fills in for the oncoming one.

Hang time: the digger waiting, bucket full, for a truck.

High-reliability organisation (HRO): an organisation that does dangerous work every day with very few serious failures, because of how its people think and act on every shift.

Leading hand: an experienced operator who runs an area or a task for the supervisor.

Level of work: the time span of the longest task a role is accountable for (Jaques; Systems Leadership). The Supervisor is level II: 3 months to 1 year. On shift, your focus is this shift and the next.

Load time: spot plus load, truck stopped to last bucket.

Lookout: a point above the pit where you can see the whole circuit; the touch-base is held there.

Mud map: a simple drawing of the pit for the shift: where each digger is, the roads, the dumps and the hazards.

Number (the): the loads you commit to for the shift, at the Drive-around after Block 1, having seen the ground. Reported in loads and buckets per hour.

Terms, O to Z

Operator scorecard: each operator's trend over the roster, on the wall in the pre-start room.

Operator's count card: the card each operator writes their own loads and buckets on, by hand, every block.

PADAL: the five numbers your shift is measured on. See "How your performance is measured: PADAL" in chapter 4.

Part B: the second part of this program, after Part A (this book). It covers optimal mining conditions, and how your contract pays, in more detail.

Pass: one bucket into a truck. Pass match: the number of passes it takes to fill the truck.

PAVA: Plan, Actual, Variance/Impact, Actions taken: how every number is shown.

Payload: what one truck carries, against its target.

Pit map: a map of the pit showing the faces, benches, roads and dumps. With the mud map, it is how the pit is checked at the handover.

Primary work time: the hours the digger is loading; the only time revenue is made.

Prime, Validation Lab, Apply, Verify: the four steps of the program: your shift as it is today, one day in the room, the shift run to the cards, one observed shift.

Productivity levers: the two things that move total volume, operating time and rate, and what sits under each (chapter 4).

Queue time: a truck waiting at the face for the one in front.

ROM: run of mine: ore or coal as mined, before processing; the pad it is delivered to.

Safety interaction: a short, two-way talk with an operator at the work about the job in front of them.

Secondary work time: time a machine is working but not loading, such as walking between faces.

Shift change: the gap between one crew's last bucket and the next crew's first.

SMU: service meter unit: the engine-hour meter on a machine. SMU hours are the hours the engine ran.

System (the): the site's electronic record (payloads, cycle times) where it has one. Its figure sits beside yours.

Spot: a truck backing into position at the digger.

TARP: trigger action response plan: what to do, set in advance, when a limit is reached.

Time in field: the four blocks, in the pit: at least 90% of your shift.

Time in office: the Shift Report and other admin, done in Block 4 and kept short.

Tip head: the edge of the dump where trucks tip.

Toe: rock left at the bottom of the face.

Touch-base 11:30: 15 minutes with the Superintendent, face to face, at the lookout on day shift: Block 2 in numbers and a decision for each area.

Truck allocation: which trucks, and how many, run on each circuit.

Utilisation: the share of available time the gear is really working, not waiting, walking or parked.

What we got away with: a weak signal: something that nearly went wrong and did not, this time.

Windrow: the bank of rock along the edge of a road or a dump.

The evidence behind the model

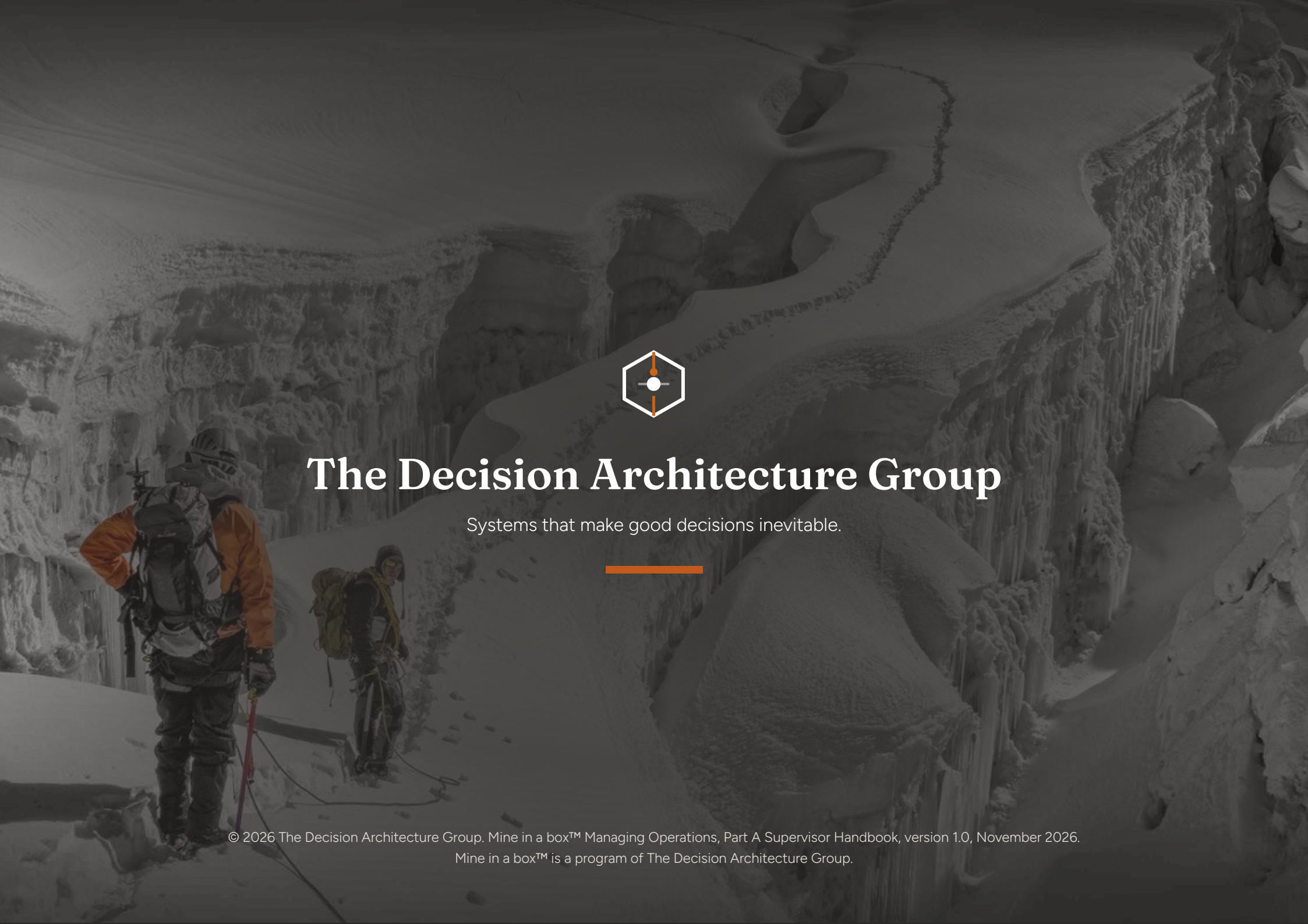
The way the shift is run here rests on a few well-tested ideas.

- **What the supervisor checks, the crew does (Zohar 2002; Zohar and Luria 2003).** Supervisors were given feedback on how often they talked with their people about safety at the work. Those talks went up, safe behaviour rose and injuries fell, with no new rules. That is why there is a safety interaction at the work every shift.
- **A decision made under pressure depends on who makes it (Kahneman, Sibony and Sunstein 2021).** Different people give different answers to the same problem, and so does one person on different days. A rule or contingency agreed before the work takes that out.
- **Small failures warn of big ones (Weick and Sutcliffe 2015).** High-reliability organisations work to five principles: preoccupation with failure, sensitivity to operations, reluctance to simplify, deference to expertise and commitment to resilience. They look hard at what nearly went wrong. That is why what we got away with is written down every shift, and why your shift runs to the five HRO questions.
- **Every role has a time span (Jaques 1989; Macdonald, Burke and Stewart 2018).** Requisite organisation and Systems Leadership set each level of work by the time span of its longest task. That is why your job and your Superintendent's job are different jobs, not bigger and smaller versions of one (chapter 2).
- **The best managers watch the work (Komaki 1986, Journal of Applied Psychology; Komaki, Desselles and Bowman 1989).** What set the most effective managers apart was time spent watching the work, so their feedback matched what people had really done. That is why you spend 90% of your shift in the field (chapter 2, the six behaviours).
- **People learn by doing, not by being told (Burke and others 2006).** Hands-on, two-way training gave about three times the learning of talks and handouts, and fewer incidents. That is why you learn on your own shift, from the cards.

Where it has run. Indicative results from client sites (unpublished): fewer safety incidents, more production and lower staff turnover where the full program is deployed. Results vary by site and by how fully the program is deployed.

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